

26. Answer: A

Magnetic flux density created by P, $B_P = \frac{\mu_0 N_P}{L} I$

Magnetic flux density created by Q, $B_Q = \frac{\mu_0 N_Q}{L} I$

Given that both solenoids P and Q are of the same length (L constant) and are connected in series (current flowing through the solenoid I is the same).

Since the magnetic flux density at point X is always zero, it implies $B_P = B_Q$ in magnitude.

$$\Rightarrow \frac{\mu_0 N_P}{L} I = \frac{\mu_0 N_Q}{L} I$$

$$\Rightarrow \frac{N_P}{N_Q} = 1$$